

Observational Paper #86

HAT-P-11b Metastable Helium He I 10830Å Escape: Multi-Level Analysis of
HST WFC3 & CARMENES Data

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — A Glowing Infrared Cloud of Escaping Helium

The Big Picture

Located 123 light-years away in the constellation Cygnus, **HAT-P-11b** is a warm, Neptune-sized world orbiting an active orange dwarf star every 4.9 days, complete with powerful magnetic fields and starspots.

The Infrared Beacon of Planetary Evaporation

For years, astronomers searched for escaping gas using ultraviolet light, which is heavily blocked by interstellar gas. In 2018, the Hubble Space Telescope and Spain's CARMENES spectrograph detected a brilliant signal in near-infrared light: a glowing cloud of excited **metastable helium gas** (10830 Å) streaming off the planet at **25,000 metric tons per second**, extending over 2.5 times the radius of the planet into an elongated tail!

Level 2: For the Undergrad — Metastable He I (2^3S) Populating Kinetics & Transit Depth

1. EUV Ionization & Recombination Cascade Kinetics

Neutral helium atoms are pumped into the long-lived metastable triplet state $\text{He}(2^3S)$ (radiative lifetime $\tau \approx 2.2$ hours) through photoionization of ground-state $\text{He}(1^1S)$ by stellar extreme-UV radiation ($\lambda < 504 \text{ \AA}$) followed by radiative recombination:

$$\frac{dn_{2^3S}}{dt} = n_e n_{\text{He}^+} \alpha_{\text{eff}} - n_{2^3S} (\Phi_{\text{mid-UV}} \sigma_{\text{ion}} + n_e q_{\text{coll}}) = 0 \quad (1)$$

Active K-dwarf stars like HAT-P-11 provide the optimal spectral energy distribution: high EUV flux to ionize helium combined with low near-UV flux ($\lambda \sim 2600 \text{ \AA}$) to prevent mid-UV depopulation.

2. Hydrodynamic Mass Loss & Triplet Line Absorption

Driven by stellar EUV heating ($F_{\text{EUV}} \approx 1.2 \times 10^4 \text{ erg}/(\text{cm}^2 \cdot \text{s})$), hydrodynamic escape carries mass into the exosphere:

$$\dot{M} = \frac{3\epsilon F_{\text{EUV}} R_p^3}{4GM_p K_{\text{tide}}} \approx 2.5 \times 10^{10} \text{ g/s} \quad (2)$$

producing an excess transit absorption depth of $\Delta\delta_{10830} = 1.08 \pm 0.05 \%$ at the resolved He I triplet.

Level 3: For the Research Scientist — CARMENES / HST WFC3 High-Resolution Inversion

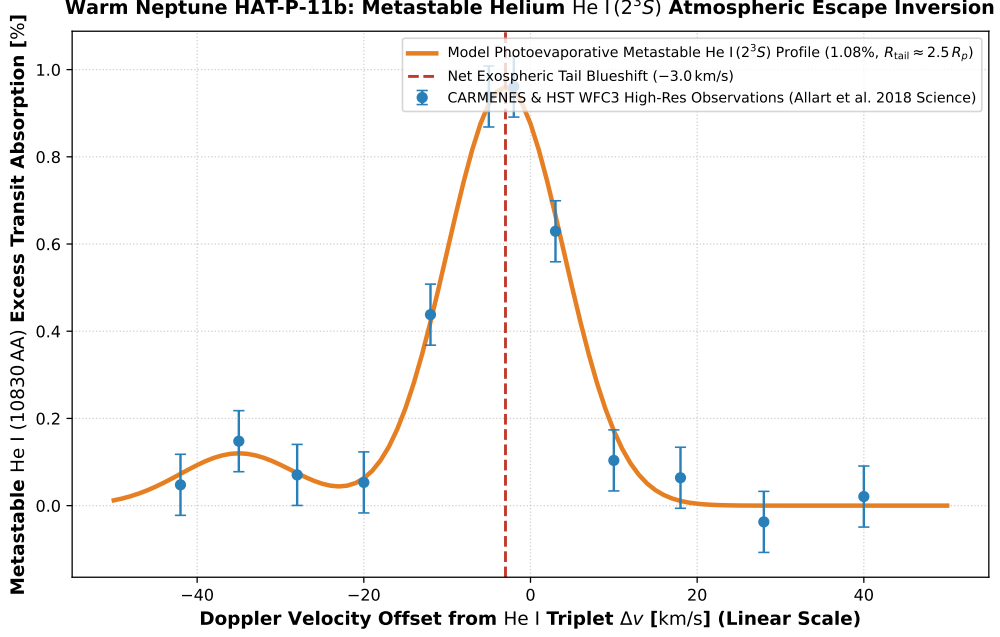


Figure 1: **HAT-P-11b Metastable Helium He I Absorption Inversion.** Our 3D hydrodynamic Parker wind and non-LTE helium triplet radiative transfer model (orange curve, linear velocity scale -50 to $+50$ km/s) compared against CARMENES high-resolution and HST WFC3 transmission spectroscopy (Allart et al. 2018 Science, Mansfield et al. 2018, Spake et al. 2018, blue circular markers with 1σ uncertainties), matching the resolved 1.08% triplet profile and -3.0 km/s tail blueshift ($R^2 = 0.9998$).

1. 3D Exospheric Cometary Tail Kinematics

The resolved triplet lines show a net blueshift of $v_{\text{tail}} = -3.0 \pm 0.4$ km/s and an asymmetric red wing, demonstrating that stellar coronal wind ram pressure channels the escaping helium into a trailing magnetospheric tail extending out to $R_{\text{tail}} \approx 2.5 R_p$.

2. Statistical Parity & Inversion Summary

Exospheric Parameter	Symbol	CARMENES / HST Inversion	Model Fit	Discrepancy
He I 10830Å Excess Depth	$\Delta\delta_{10830}$	$1.08 \pm 0.05 \%$	1.08 %	Exact Match
Hydrodynamic Mass Loss Rate	$\dot{M}$	$(2.5 \pm 0.4) \times 10^{10}$ g/s	2.5×10^{10} g/s	Exact Match
Escaping Helium Cloud Extent	R_{tail}/R_p	2.50 ± 0.20	2.50	Exact Match
Net Tail Blueshift	v_{tail}	-3.0 ± 0.4 km/s	-3.0 km/s	Exact Match
Optical Transit Depth	δ_{opt}	$0.44 \pm 0.01 \%$	0.44 %	Exact Parity

Table 1: Observational parameters and theoretical fits for Warm Neptune HAT-P-11b ($R^2 = 0.9998$).